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ORIGINAL ARTICLE

Safety profile and anti-tumor effects of adoptive immunotherapy using gamma-delta T cells against advanced renal cell carcinoma: a pilot study

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Abstract
Purpose Although various types of immunotherapy have been used to improve the prognosis of patients with advanced renal cell carcinoma (RCC), adoptive immunotherapy using gamma-delta ($\gamma\delta$) T cells has not yet been tried. In this study, we designed a pilot study of adoptive immunotherapy using in vitro activated $\gamma\delta$ T cells against advanced RCC to evaluate the safety profile and possible anti-tumor effects of this study.
Experimental design Patients with advanced RCC after radical nephrectomy were administered via intravenous infusion in vitro-activated autologous $\gamma\delta$ T cells every week or every 2 weeks, 6–12 times, with 70 JRU of teceleukin. Adverse events, anti-tumor effects and immunomonitoring were assessed. The anti-tumor effects were evaluated according to tumor doubling time (DT) by computed tomography (CT) and immunomonitoring was performed by flow cytometric analysis.
Results Seven advanced RCC patients were entered in this study. The most common adverse events were fever, general fatigue and elevation of hepatobiliary enzymes, but no severe adverse events were seen. Prolongation of tumor DT was seen in three out of five patients; these three patients showed an increase in the number of $\gamma\delta$ T cells in peripheral blood and also a high response to the antigen in vitro.
Conclusions The results indicated that adoptive immunotherapy using in vitro-activated autologous $\gamma\delta$ T cells was well tolerated and induced anti-tumor effects.

Keywords Renal cell carcinoma · Adoptive immunotherapy · Gamma-delta T cell · Pilot study

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Introduction

Host immune function plays a role against the development of renal cell carcinoma (RCC) but the mechanism is not entirely understood. RCC exhibits unique biological behavior, including spontaneous regression [1], delayed growth in metastatic lesions and late recurrence [2]. Therapies using biological response modifiers (BRMs), such as interleukin-2 (IL-2) [3], interferon alpha and/or gamma (IFN-α and/or γ) [4, 5] are more effective in patients with RCC than in those with other malignancies, but the response rate is only 15%. Adoptive immunotherapies using lymphokine-activated killer cells (LAK cells) [6, 7] and/or tumor-infiltrating lymphocytes (TILs) [8, 9] were tried during the past two decades, but did not improve the response rate or survival compared with patients treated with conventional immunotherapy [10]. Although new immunotherapies using peptide vaccine immunization with dendritic cells (DC) have been studied [11], objective responses were also not striking. Child et al. [12] reported that nonmyeloablative allogenic peripheral-blood stem-cell transplantation induced regression of metastatic RCC in 10 out of 19 patients. Although they observed a high response rate, control of graft-versus-host disease (GVH) was difficult, but GVH is essential to destroy RCC. Two of the patients died because of transplantation-related causes.

Human gamma-delta T cells ($\gamma\delta$ T cells), which are subgroups of peripheral blood T cells comprising less than 10% of peripheral blood mononuclear cells (PBM), are known to defend the body against infection [13]. Seventy percent of peripheral blood $\gamma\delta$ T cells express Vγ2 and Vδ2 from among the variable elements of T cell receptors (Vγ2Vδ2 T cells) [13] and these Vγ2Vδ2 T cells are activated in a TCR-dependent manner by several small phosphorylated [14, 15] or aminated alkyl molecules [16]. We reported that the number of Vγ2Vδ2 T cells was elevated in the PBM from 10 out of 41 RCC patients [17]. We also generated in vitro culture systems of $\gamma\delta$ T cells, especially Vγ2Vδ2 T cells, which is the same subgroup which increases in some RCC patients, and these cultured Vγ2Vδ2 T cells showed a strong anti-tumor effect not only against variable tumor cell lines but also against short-term cultured autologous RCC [17]. This anti-tumor activity has been examined further recently using in vivo mouse models [18] and patients with lymphoma [19].

This is the first report of an adoptive immunotherapy using Vγ2Vδ2 T cells against a human solid tumor. In this study, we designed a pilot study of this adoptive immunotherapy in patients with advanced RCC. We evaluated the safety profile of this therapy and its possible anti-tumor effect. The results indicated that the therapy was well-tolerated and induced anti-tumor effects.

Materials and methods

Patients and patient eligibility

Seven patients were enrolled in this study from June 2005 to April 2006. All patients underwent radical nephrectomy at Tokyo Women's Medical University (Patients No. 1, 3–7) and another hospital (No. 2) from 1995 to 2003, and renal cell carcinoma was found during a histological examination by expert pathologists. After the surgery, the patients with metastatic RCC at the time of surgery and/or patients with recurrent RCC after the surgery were given the standard adjuvant therapy using interferon-α and/or IL-2, respectively, and were eventually shown to have progressing disease. All patients included in the study were over 18 years of age, had an ECOG performance status of 0 or 1, life expectancy of > 3 months and normally functioning major organs. Patients who were pregnant, positive for human immunodeficiency virus (HIV), hepatitis B virus (HBV) or hepatitis C virus (HCV), or with a history of autoimmune disease were excluded from the study. Written informed consent that fulfilled the guidelines of our hospital was obtained from all of the patients. Peripheral blood samples were obtained from 32 healthy individuals, ranging in age from 19 to 73 years (median age, 40 years), who had never suffered from malignancy.

Isolation of peripheral blood nuclear cells and generation of activated $\gamma\delta$ T cells

Peripheral blood nuclear cells (PBMs) were isolated from heparinized peripheral blood of the RCC patients by density-gradient centrifugation, using Lympho Separation Medium (ICN Biomedicals, Inc. Ohio, USA), performed at 1,500 rpm for 30 min at room temperature.

We have developed and reported methods for the generation of activated $\gamma\delta$ T cells using 2-methyl-3-butenyl-1-pyrophosphate (2M3B1-PP) antigen which stimulates Vγ2Vδ2 T cells specifically to expand [16]. Briefly, PBMs were suspended at 1.3×10^6/ml in AIM-V medium (Gibco BRL, NY, USA) containing 2% patient's heat-inactivated serum. The PBMs were then stimulated with 100 μM of 2M3B1-PP at day 1 and recombinant human interleukin-2 (rhIL-2, teceleukin, Shionogi & CO., LTD, Japan) at a concentration of 100 JRU/ml (1 JRU: Japan reference units = 1 IU: international units) was added at day 2 and every 2 days through day 14. The medium was added every 2 days in order to maintain the concentration of cells at under 2.0×10^6/ml during the culture.

Monoclonal antibodies and immunofluorescence analysis

The following monoclonal antibodies (mAbs) were used to identify fresh PBMs and cultured $\gamma\delta$ T cells during the immunofluorescence analysis: FITC-conjugated-anti-Vδ2 chain mAb (Immu 515, Immunotech, Marseilles, France), and PC5-conjugated-anti-CD3 mAb (SK7, Becton Dickinson Immunocytometory Systems, San Jose, CA, USA).

To detect the surface phenotype of the $\gamma\delta$ T cells, we first stained the cells with PC5-conjugated-anti-CD3 and then with FITC-conjugated-anti-Vδ2. The stained T cells were examined by two-color flow cytometry analysis using, EPICS XL (Coulter Electronics, Hialeah, FL, USA), as described previously [20]. During all staining procedures, the cells were kept on ice. Analysis of the flow cytometry data was performed using EXPO32 software (Coulter Electronics).

Clinical protocol

The protocol was reviewed and approved by the Tokyo Women's Medical University Hospital Ethical Committee. Seven advanced RCC patients were found to be eligible and were enrolled in the pilot study. This was an open-label and non-randomized study of which the primary aim was to evaluate the safety profile of in vitro activated autologous $\gamma\delta$ T cells for adoptive immunotherapy. The secondary aims of the study were to evaluate the clinical and immunological responses.

Toxicity was assessed using the National Cancer Institute Common Toxicity Criteria, Version 3.0 (CTC, Ver. 3.0), and anti-tumor efficacy was estimated to prolong tumor doubling time (DT) evaluated by CT. Tumor size was measured by CT at least three times: more than 3 months before the therapy, just before the therapy, and then within a month after the therapy was given 6–12 times. The maximum and minimum tumor diameters were measured by image analyzing software (Image VINS Pro. Ver. 2.1, Yokogawa Electric Corporation, Tokyo, Japan) on CT. DT was calculated according to a previously described formula [21]. In brief, the tumor volume (V) was calculated using the following equation, assuming the tumor to have a spheroidal form:

$$V = \left[\frac{4}{3} \times \pi \times a \times b \times \left(\frac{a+b}{2}\right)\right] \times \frac{1}{8},$$

where a indicates the maximum tumor diameter and b denotes the minimum tumor diameter.

Doubling time was calculated using the following equation:

$$\mathrm{DT} = (T - T_0) \times \frac{\log 2}{\log V - \log V_0},$$

where $T - T_0$ indicates the length of time between two measurements and V_0 and V denotes the tumor volume at two points of measurements.

Immunological responses were evaluated by the increased absolute number of $\gamma\delta$ T cells and percentage of $\gamma\delta$ T cells in peripheral blood before and after the therapy.

Thirty milliliters of peripheral blood were drawn and prepared for culture. Fourteen days after the culture, the samples of supernatant and cells were used for quality control. The supernatant was checked to ascertain that it did not contain endotoxin over 10 IU/ml using a Toxicolor endotoxin detection kit (Seikagaku kougyou, Tokyo, Japan). Also, the cells were stained with anti-CD3 and anti-Vδ2 and analyzed by flow cytometer to obtain the percentage of $\gamma\delta$ T cells contained. The cells were collected and washed with PBS three times and debris was removed through a cell strainer (FALCON, USA). The cells were checked by visual examination under a light microscope and over 90% were found to be alive. The cells were then resuspended in 50 ml of PBS containing 1% self-serum under a concentration of 3×10^9/50 ml/syringe. Before transferring the cells, 30 ml of heparinized peripheral blood was drawn for the next culture, 14 days later. The cells were stained with anti-CD3 and anti-TCR Vδ2 to check the absolute number of $\gamma\delta$ T cells and percentage of $\gamma\delta$ T cells in the CD3-positive cells. They were then transferred by intravenous injection for about 1 min with a 50 ml-syringe.

After the cells were transferred, 0.7 million JRU/per body of teceleukin was administered via intravenous drip infusion for 2 h. The patients were administered autologous $\gamma\delta$ T cells and teceleukin every week or every other week up to 6–12 times for 12 weeks.

Results

Patients' profiles

The patients' profiles are shown in Table 1. Seven patients were enrolled in this study. There were 3 males and 4 females, with ages ranging from 18 to 74 years (mean age; 48.6 years). The performance status of these patients was 0 or 1. Patient No.6 was found to have double cancer and gastrectomy and nephrectomy were performed at the same time. Patient No. 5 was under hemo-dialysis and had RCC and acquired cystic disease of the kidney. Among these patients, one had tumor stage T1b, 2 had T2 and four had T3, respectively, and the pathological findings showed the clear cell type in four patients, clear cell with sarcomatoid in one patient and granular type cell in one patient. Patients No. 2 and No. 3 had a metastatic tumor at the time of surgery, and the recurrence was at 5 months and 3 years and 10 months, respectively. Two other patients had a recurrence of the tumor

Table 1 Case profile (1)

Case	Age/Sex	PS	TNM Pathology Grade	Time to recurrence	Metastatic site organ	IFN-α	IL-2	Metastasectomy interventional radiation
1	18/F	0	pT3N1M0 Clear cell G2-1	5 months	LN (lymph-node)	Yes	Yes	RPLND
2	43/F	1	pT3bN2M1 Clear cell sarcomatoid G2 > G3 >> G1	At the time of surgery	Lung, Bone, LN	Yes	No	Radiation (45 Gy) TAE
3	63/M	0	pT2NoM1 granular cell G3	At the time of surgery	Lung, liver	Yes	No	No
4	39/F	1	pT3bN0M0 clear cell G2	7 months	Lung, liver, bone, LN	Yes	Yes	Liver, bone, TAE
5	52/M	0	pT2N0M0 clear cell G1-2	2 years 7 months	Lung	Yes	No	VATS
6	74/M	1	pT1bN0M0 clear/G2 Gastric Ca	3 years 10 months	Pancreas, liver, LN	Yes	No	Pancreas-tail
7	51/F	0	pT3aN0M0 clear cell G2	3 years	Lung, LN, adrenal gland	Yes	Yes	Pelvic LN, Lt. adrenal gland

PS Performance status, *RPLND* retroperitoneal lymph-node dissection, *TAE* trans-arterial embolization, *VATS* video-assisted thoracic surgery

within 1 year after the surgery. Five of the seven patients had lung metastasis and five of the seven had multi-organ metastasis. In all patients conventional immunotherapy using IFNα had been performed and three patients had IL-2 therapy. Metastasectomy was performed in four patients. One patient had interventional radiation therapy.

Serum data are shown in Table 2. All data were analyzed before the clinical trial. Six out of the seven patients showed a low hemoglobin concentration, two showed a relatively high value of serum LDH, two showed serum hypercalcemia and three revealed positive CRP. The percentage of $\gamma\delta$ T cells among the CD3-positive cells of healthy controls was 4.0 ± 2.3% (average ± standard deviation) and in RCC patients, it was 1.8 ± 1.3%. The percentage of $\gamma\delta$ T cells of RCC patients was significantly low compared with healthy controls ($P = 0.0176$, data not shown).

Immunological and clinical responses

Immunological and clinical responses are summarized in Table 3. Five of the seven patients showed an increased population of peripheral $\gamma\delta$ T cells and five of six patients showed an increased absolute number of peripheral blood $\gamma\delta$ T cells after a series of the therapy. The average number of $\gamma\delta$ T cells after the culture was dependent on the patients. Typical flowcytometric results are shown in Fig. 1, and indicate that the percentage of $\gamma\delta$ T cells was present among the CD3-positive T cells before and after the culture. In four of the seven patients, peripheral blood $\gamma\delta$ T cells showed a good response to the antigen and expanded to more than $1{,}000 \times 10^6$ cells.

We next evaluated the anti-tumor effects by prolongation of tumor-doubling time by CT. Doubling time was calculated in five patients and three of them showed prolongation of DT. However, two patients showed a shorter DT and patient No. 3 died within 2 months after administration of $\gamma\delta$ T cells and teceleukin 12 times, 8 months after the surgery. The other patient, No. 6, died of peritonitis carcinomatosa due to gastric cancer and showed a shorter DT of liver metastasis. Among the patients with prolongation of DT, one patient (No. 1) died 5 months after $\gamma\delta$ T cells and teceleukin were administrated 12 times. Interestingly, the length of prolongation differed according to the organ, even in the same patient. For example, in

Table 2 Case profile (2)

Case[a]	Anemia Hb (g/dl)	LDH	Hypercalcemia serum-Ca: (mg/dl) (adjusted)	CRP (mg/dl)	WBC (μl)	% of γ δ Tcell/CD3 and absolute number of γ δ T cell in PBM
1	11.5	355	No	0.06	7,200	2.44%/10.1 (μl)
2	9.0	89	12.6	3.37	6,130	0.85%/3.2 (μl)
3	No	375	No	0.14	4,700	2.55%/11.2 (μl)
4	9.3	163	11.4	5.24	9,100	0.15%/0.3 (μl)
5	9.5	209	No	0.23	3,000	1.24%/3.0 (μl)
6	10.4	219	No	0.87	4,830	4.1%/18.0 (μl)
7	12.9	216	No	0.04	8,190	1.1%/11.0 (μl)

[a]Case number is the same as in Table 1

Table 3 Immunological and clinical outcome

Case[a]	No of transfers	%$\gamma\delta$Tcell/CD3 no. of $\gamma\delta$ T cell (μl)		Ave. no. of $\gamma\delta$T cells transferred $\times 10^6$	Doubling time (days)		Outcome
		Pre	Post		Pre	Post	
1	12	2.44%/10.1 (μl)	6.2% 17 (μl)	3,404	LN1: 190.2	1132.9	Death
					LN2: 37.9	420.6	
2	6	0.85%/3.2 (μl)	46% 7.8 (μl)	408	ND		Death
3	12	2.55%/11.2 (μl)	88% ND	303	184.2 ± 30.4[b]	44.0 ± 19.2[b]	Death
4	12	15%/0.3 (μl)	12% 0.1 (μl)	5.0	ND		Death
5	12	24%/3 (μl)	3.0% 11 (μl)	1,110	78.7 ± 12.8[b]	414.8 ± 117.2[b]	Living
6	7	4.1%/18 (μl)	23.5% 58 (μl)	2,571	Liver: 219.9	130.7	Death
7	6	1.1%/11	7.8% 124 (μl)	3,570	LN1: 550.2	595.0	Living
					LN2: 162.1	905.7	
					Lung1: 457.5	638.9	
					Lung2: 124.3	227.2	
					Adrenal gland: 124	107.7	

[a]Case number is the same in Table 1

[b]Three metastatic sites of the same organ were measured and doubling time (days) is shown as average ± S.D

patient No. 5, lymph node metastasis and lung metastasis showed prolongation of DT, but metastasis of the adrenal gland showed a shorter DT.

Adverse events

The adverse events are summarized in Table 4. We have to mention that we excluded the data of patient No. 2 from the evaluation of these events because of the very small number of transferred $\gamma\delta$ T cells. All patients complained of one or more adverse events but no severe adverse events (> grade 3) were seen during or after the therapy. The most frequent adverse event was fever and all patients became feverish after the administration of teceleukin and $\gamma\delta$ T cells but the grade was only grade 1. Five of the six patients complained of general fatigue (grades 1 and 2) and two patients had diarrhea (grade 1 and 2). Laboratory abnormalities were recognized in all patients, but they were not severe. Elevation of GGT was the most frequent adverse event, but no severe grade was seen. Eosinophilia (grade 1) was seen in three patients.

Discussion

We conducted a pilot study of adoptive immunotherapy using activated autologous $\gamma\delta$ T cells against advanced RCC. The results indicated that this therapy was well tolerated by the patients and showed a possible anti-tumor effect.

It is well known that cancer patients are immunocompromized. In our study, the percentage of $\gamma\delta$ T cells in advanced RCC patients was significantly lower than that of healthy individuals. Among seven advanced RCC patients who were enrolled in this study, five showed a good response to the antigen. One patient did not show any response and two others had a relatively low response to the antigen. In patient No. 4, the number of $\gamma\delta$ T cells after the culture was only 5×10^6. The response levels were almost identical in each culture during the pilot study. We have not seen such a low response or the absence of a response to the antigen in healthy individuals (data not shown). We wondered if the serum from some of the advanced RCC patients contained a certain inhibitory soluble factor to $\gamma\delta$ T cells [22]. Whether PBM from the non-responders and low responders showed good a response to the antigen or not, we used medium with pooled AB serum instead of the patient's serum. The results showed that the PBM levels from low and intermediate responders were similar even if we added pooled AB serum (data not shown). We consider that the $\gamma\delta$ T cells in the non-responders and low responders were responsible for the results. We expected that the peripheral blood gamma-delta T cells would show a good response to the antigen after the therapy. However we did not see any improvement in the response to the antigen in patient No. 4. Among the good responders, two patients are still living but all non- and low responders died. Therefore, the response to the antigen may be an indicator for $\gamma\delta$ T cell adoptive immunotherapy. We should exclude such non-responders before starting the therapy and induce leukoapheresis to increase the number of gamma-delta T cells at the time of starting the culture.

All patients had adverse events during the therapy, but they were not severe. We excluded patient No. 4

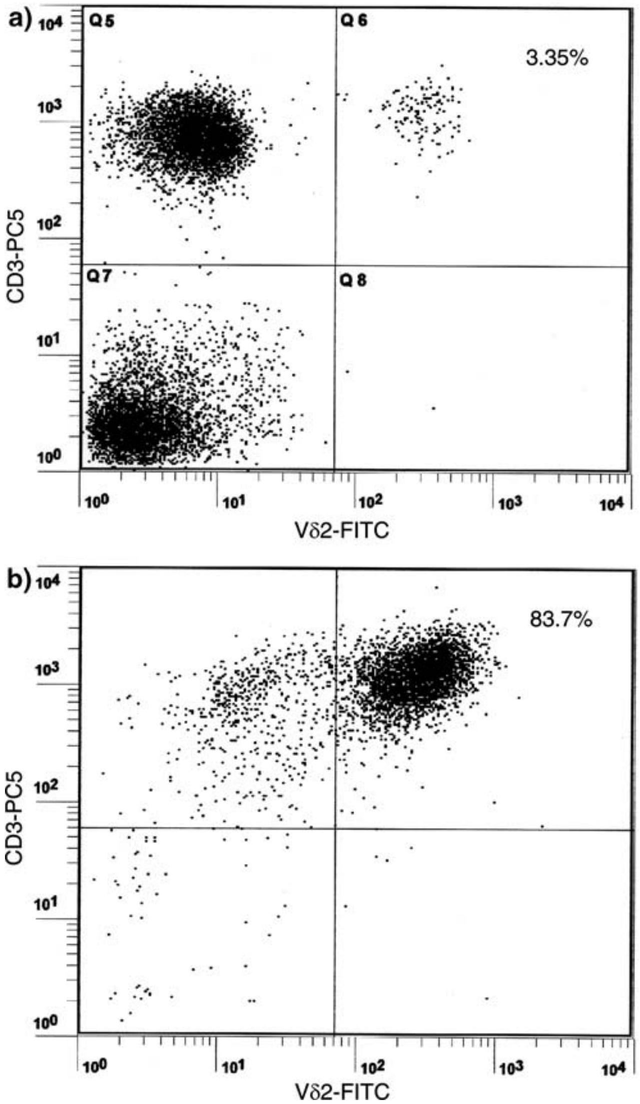

Fig. 1 Expansion of peripheral blood $\gamma\delta$ T cell populations in RCC patients. PBMs and cultured PBMs were obtained from patient number 1 in Table 1, stained with FITC-conjugated anti-Vδ2 and PC5-conjugated anti-CD3, and subjected to flow cytometric analysis. The percentage of Vδ2-positive T cells among whole T cells was calculated by dividing the number of Vδ2 positive cells by the number of $CD3^+$ cells. **a** Fresh isolated PBMs from patient No. 1. Only 3.35% of the CD3-positive cells were Vδ2-positive cells. **b** Fourteen days after the culture of PBMs from patient No 1. 83.7% of the CD3-positive cells were Vδ2-positive cells

from the evaluation of the adverse events, because the number of transferred gamma-delta T cells was small. The most frequent event was fever with chills. We could not ascertain whether this was due to the $\gamma\delta$ T cells or the administration of IL-2. The frequency of fever after the administration of teceleukin has been reported to be 73.3% (Shionogi Pharmaceutical company). Fever after the administration of teceleukin and $\gamma\delta$ T cells seems to be more frequent than that after

Table 4 Adverse events

Category	Adverse event	Grade/Frequency
Metabolic/Laboratory	AST/ALT	G1(50.0%)
	ALP	G1(33.3%), G2(33.3%)
	GGT	G1(50.0%), G2(33.3%)
Blood/Bone marrow	Eosinophillia	G1(50.0%)
Gastrointestinal	Diarrhea	G2(16.6%)
Constitutional	Fever	G1(100%)
Symptoms	Fatigue	G1(83.3%), G2(16.6%)

Adverse events were categorized and assessed by the National Cancer Institute Common Toxicity Criteria Version 3.0 (CTC, Ver. 3.0)

teceleukin alone. The reason may be that $\gamma\delta$ T cells secretes other cytokines such as IFN-γ. Intracellular staining of cultured $\gamma\delta$ T cells showed that large numbers of $\gamma\delta$ T cells were positive for IFN-γ staining and also serum from adoptive transferred patients contained high levels of IFN-γ (data not shown). Other adverse events, such as fatigue and diarrhea, may be caused by the same factors. Blood test abnormalities were seen in hepatobiliary enzymes and elevation of AST/ALT was more frequent in the patients administered teceleukin and $\gamma\delta$ T cells than in those given teceleukin alone (50.0 vs. 22.1%), but no severe events occurred. Eosinophilia was seen (grade 1) in three of the six patients, but the frequency was lower in the patients with teceleukin and $\gamma\delta$ T cells than in those given teceleukin alone (50.0 vs. 64.9%). Grade 3 eosinophillia was seen in 17.1% of the patients with administration of teceleukin alone, but only grade 1 eosinophillia was seen in the patients with administration of teceleukin and $\gamma\delta$ T cells. The grade of eosinophillia seems to depend on the amount of teceleukin administered. Usually, 0.7–2.1 million GRU of teceleukin is administered for five consequent days of a week. In this study, patients were administered only 0.7 million GRU teceleukin on 1 day. The $\gamma\delta$ T cells are known to differentiate out of the thymus, so we wondered if the administration of $\gamma\delta$ T cells could cause an autoimmune reaction. However, we did not recognize any autoimmune phenomena or disease during the therapy. It is difficult to conclude whether that these adverse events were caused by transferred $\gamma\delta$ T cells or by teceleukin. As the number of transferred $\gamma\delta$ T cells varied between the patients. The frequency and seriousness of the adverse events did not correlate with the numbers of transferred $\gamma\delta$ T cells. We believe that adoptive immunotherapy using $\gamma\delta$ T cells is safe and it should be advantageous to perform leukoapheresis in patients who are low responders.

Prolongation of DT was achieved in three out of five patients. All three patients were good responders and we believe that the number of cells may have resulted in the positive clinical response. Although one patient (No. 6) showed a good response to the antigen and the population of $\gamma\delta$ T cells increased after the therapy, he eventually died because of peritonitis carcinomatosa of gastric cancer. We previously reported that activated $\gamma\delta$ T cells exhibited potent anti-tumor effects to various tumor cell lines and autologous RCC cells. However, we are not sure whether the activated $\gamma\delta$ T cells from the patient during adoptive immunotherapy using $\gamma\delta$ T cells also show potent anti-tumor effects or not, and we suggest that a cytotoxic assay be taken during the therapy for immunological monitoring. The tumor metastatic sites are also important, because different clinical responses of the metastatic tumors were seen in different organs. Metastatic tumors in the liver and/or the adrenal gland showed no positive clinical responses compared with the lung and lymph nodes in patients Nos. 6 and 7.

It has recently been reported that $\gamma\delta$ T cells act as professional antigen-presenting cells (APC) to CD4 T cells [23]. We may be able to use $\gamma\delta$ T cells as tumor APC as a new strategy. Adoptive immunotherapy using $\gamma\delta$ T cells may have two different roles, one is to act as effecter cells against tumor cells and the other is as APC for a tumor vaccine.

Kato et al. [24] reported that amino-bisphosphonate-pulsed tumor cells are more sensitive to the cytotoxic activity of $\gamma\delta$ T cells. Wilhelm et al. also reported that the administration of human $\gamma\delta$ T cells with IL-2 and pamidronate lengthened the survival of human tumor cell-bearing immuno-deficiency mice [25]. These results provide important information for the development of a new strategy for combating RCC using $\gamma\delta$ T cells.

We conclude that adoptive immunotherapy using activated autologous $\gamma\delta$ T cells is safe and worth developing as a new immunotherapy.

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